

M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 9 (9TH MATRIC) - PHYSICS

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Science Group | Total Marks: 75 Marks (Theory: 60 | Practical: 15) | Time: 2 Hours 45 Minutes

SECTION 1: BISE GUJRWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- Section A: Objective (MCQs):

12 Compulsory MCQs (1 Mark each = 12 Marks). 1-2 questions per unit testing definitions, SI units, and standard formulas.

- Section B: Short Questions:

Attempt 15 out of 24 Short Questions divided into 3 sub-parts (Q2, Q3, Q4: 5/8 each = 30 Marks). Focus on scientific laws, derivations, and reasoning.

- Section C: Long / Numerical Questions:

Attempt 2 out of 3 Long Questions (Q5, Q6, Q7: Part (a) Theory 5 Marks + Part (b) Numerical 4 Marks = 18 Marks).

- SLO Cognitive Distribution:

Knowledge: 50% | Understanding: 35% | Application & Numerical Analysis: 15%.

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Unit 1: Physical Quantities & Measurement:

Base vs Derived Quantities, 7 SI base units, Prefixes (micro, nano, pico), Scientific notation, Least count of Vernier Calipers (0.1 mm / 0.01 cm) and Screw Gauge (0.01 mm / 0.001 cm), Zero error correction, Significant figures rules.

Unit 2: Kinematics:

Rest vs Motion, Translatory, Rotatory, and Vibratory motion, Scalars vs Vectors, Distance vs Displacement, Speed vs Velocity, Acceleration, Derivations of 3 Equations of Motion ($v_f = v_i + at$; $S = v_i t + \frac{1}{2} a t^2$; $2aS = v_f^2 - v_i^2$), Motion under gravity ($g = 9.8 \text{ m/s}^2$).

Unit 3: Dynamics:

Force, Inertia, Momentum ($p = mv$), Newton's 3 Laws of Motion, Mass vs Weight ($W = mg$), Law of Conservation of Momentum ($m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$), Friction (static vs kinetic, advantages & reducing methods), Centripetal force ($F_c = mv^2/r$) and Banking of roads.

Unit 4: Turning Effect of Forces:

Like and Unlike Parallel Forces, Head to Tail Rule, Resolution of Vectors ($F_x = F \cos \theta$, $F_y = F \sin \theta$), Torque ($\tau = F \times L$), Principle of Moments, Centre of Mass vs Centre of Gravity, Couple, Equilibrium (1st Condition: $\sum F = 0$, 2nd Condition: $\sum \tau = 0$), States of Equilibrium (Stable, Unstable, Neutral).

Unit 5: Gravitation:

Newton's Law of Universal Gravitation ($F = G \frac{m_1 m_2}{r^2}$), Determination of Mass of Earth ($M_e = \frac{g R^2}{G}$ approx $6.0 \times 10^{24} \text{ kg}$), Variation of 'g' with altitude, Artificial satellites and Orbital velocity formula ($v_o = \sqrt{gR}$).

Unit 6: Work and Energy:

Work ($W = F \cdot S \cos \theta$, Unit: Joule), Kinetic Energy ($KE = \frac{1}{2} m v^2$), Potential Energy ($PE = mgh$), Major Forms of Energy, Interconversion of Energy, Efficiency = $(\text{Output/Input}) \times 100\%$, Power ($P = W/t$, Unit: Watt, 1 hp = 746 W).

Unit 7: Properties of Matter:

Kinetic Molecular Model, Density ($\rho = m/V$), Pressure ($P = F/A$, Unit: Pascal), Atmospheric pressure (barometer), Pascal Law (hydraulic press: $F_2/A_2 = F_1/A_1$), Archimedes Principle (Upthrust = $\rho \cdot g \cdot V$), Principle of Flotation, Elasticity and Hooke Law (Stress/Strain = Young Modulus Y).

Unit 8: Thermal Properties of Matter:

Temperature vs Heat, Thermometers, Specific Heat Capacity ($c = Q / (m \cdot \Delta T)$), Heat of Fusion (H_f) and Vaporization (H_v), Evaporation and cooling effect, Thermal expansion of solids and liquids (Linear expansion: $\Delta L = \alpha \cdot L_0 \cdot \Delta T$, Volume expansion: $\Delta V = \beta \cdot V_0 \cdot \Delta T$, where $\beta = 3 \cdot \alpha$).

Unit 9: Transfer of Heat:

Conduction (thermal conductivity k), Convection (convection currents in coastal breeze), Radiation (Stefan-Boltzmann law, black bodies as best absorbers/emitters), Applications of greenhouse effect, Thermos flask design.

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

- * Always write given data, standard formula, calculation steps, and SI units with final answers in numericals.
- * Draw neat, labeled ray and vector diagrams with a sharp pencil and ruler.
- * Memorize least counts and standard constants ($G = 6.673 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$, $M_e = 6 \times 10^{24} \text{ kg}$, $R_e = 6400 \text{ km}$).
- * Frame final answers in rectangular boxes to enable rapid scoring by board examiners.